

Predicting TV Sales using Linear Regression

*A Summer Internship Report submitted in partial fulfillment of the
requirements for the award of degree of*

BACHELOR OF TECHNOLOGY In COMPUTER SCIENCE AND ENGINEERING

**Submitted
by**

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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
ADITYA UNIVERSITY
(Formerly Aditya Engineering College (A))**

2024-2025

ADITYA UNIVERSITY

(Formerly Aditya Engineering College(A))

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the internship report entitled “**PREDICTING TV SALES USING LINEAR REGRESSION**” is being submitted by **LAGGISETTI SRIRAM SAI (22A91A05D8)** in partial fulfillment of the requirements for award of the B.Tech., degree in Computer Science and Engineering for the academic year 2024-2025.

Internship Coordinator

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DECLARATION

I hereby declare that the internship report entitled “**PREDICTING TV SALES USING LINEAR REGRESSION**” is a genuine report. This work has been submitted to the **ADITYA UNIVERSITY**, Surampalem, in partial fulfillment of the **B.Tech.**, degree. I further declare that this report has not been submitted in full or part of the award of any degree of this or any other educational institutions.

by

LAGGISETTI SRIRAM SAI

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Internship Completion Certificate



CERTIFICATE OF INTERNSHIP

This is to Certify that Mr./Ms

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Enrolled in the **Computer Science and Engineering - 22A91A05D8**

From College **Aditya Engineering College**

of university **JNTUK, Kakinada**

has Successfully Completed short-term Internship programme titled

Data Science

under SkillDzire for 2 Months. Organized By **SkillDzire** in collaboration with **Andhra Pradesh State Council of Higher Education**.

Certificate ID:
SDST-23898

Issued On:
1-Jul-2024



Approved By AICTE



Authorized Signature

ACKNOWLEDGEMENT

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It is with immense pleasure that we would like to express our indebted gratitude to our internship coordinator **A. Phani Sridhar sir, Assistant Professor**, who has guided us a lot and encouraged us in every step of the intern project work, **his** valuable moral support and guidance throughout the Intern project helped us to a greater extent.

I am grateful to **Dr. K. Swaroopa, Associate Professor and HOD** for inspiring us all the way and for arranging all the facilities and resources needed for my intern project work.

I wish to thank our **Dr. S. Rama Sree, Professor in CSE and Dean(Academics)** for their encouragement and support during the course of my intern project work.

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Not to forget, **Faculty, Lab Technicians, Non-Teaching Staff and our Friends** who have directly or helped and supported us in completing my intern project work in time.

Abstract

The primary objective of this project is to develop a predictive model using a training dataset to gain insights and make accurate predictions. The dataset, consisting of various features and labeled data points, was preprocessed to handle missing values, outliers, and normalization to ensure the quality of the input data. Feature engineering techniques were applied to extract the most relevant information, and multiple machine learning algorithms, including Decision Trees, Random Forest, and Gradient Boosting, were implemented for comparison.

The model training phase involved splitting the dataset into training and testing sets to evaluate performance metrics such as accuracy, precision, recall, and F1-score. Hyper parameter tuning was performed using cross-validation to optimize the model's performance. The final model, selected based on its predictive power and efficiency, demonstrated significant improvements in prediction accuracy over baseline methods.

This project illustrates the end-to-end process of building a data science model, from data cleaning and feature selection to model evaluation and optimization, ultimately producing a reliable tool for data-driven decision-making .

This abstract highlights the key stages of the project, from data preparation to model training and evaluation.

Learning Objectives/Internship Objectives

- Internships are generally thought of to be reserved for college students looking to gain experience in a particular field. However, a wide array of people can benefit from Training Internships in order to receive real world experience and develop their skills.
- An objective for this position should emphasize the skills you already possess in the area and your interest in learning more
- Internships are utilized in a number of different career fields, including architecture, engineering, healthcare, economics, advertising and many more.
- Some internships are used to allow individuals to perform scientific research while others are specifically designed to allow people to gain first-hand experience working.
- Utilizing internships is a great way to build your resume and develop skills that can be emphasized in your resume for future jobs. When you are applying for a Training Internship, make sure to highlight any special skills or talents that can make you stand apart from the rest of the applicants so that you have an improved chance of landing the position.
- Develop Technical Skills
- Understand ML Workflows
- Improve Problem-Solving Skills
- Enhance Data Analysis and Preprocessing Skills
- Exploring Cutting-Edge ML Technologies
- Learning Model Deployment and Optimization

WEEKLY OVERVIEW OF INTERNSHIP ACTIVITIES

1 st WEEK	DATE	DAY	NAME OF THE TOPIC/MODULE COMPLETED
	4/6/24	Tuesday	Justification of Data Science
	5/6/24	Wednesday	Introduction to Data Science
	6/6/24	Thursday	Explanation of Lists,Sets and Tuples
	7/6/24	Friday	Practice Programes Explanation
	8/6/24	Saturday	Holiday
	10/6/24	Monday	Numpy Arrays

2 nd WEEK	DATE	DAY	NAME OF THE TOPIC/MODULE COMPLETED
	11/6/24	Tuesday	Pandas Series Operations
	12/6/24	Wednesday	Pandas Aggregate Functions
	13/6/24	Thursday	SQL Introduction
	14/6/24	Friday	SQL IMDB Data Base
	15/6/24	Saturday	Holiday
	17/6/24	Monday	SQL DB

3rd WEEK	DATE	DAY	NAME OF THE TOPIC/MODULE COMPLETED
	18/6/24	Tuesday	SQL concepts-SELECT,JOINS,WHERE CLAUSE
	19/6/24	Wednesday	SQL-CASE STATEMENTS
	20/6/24	Thursday	SQL Ranking
	21/6/24	Friday	Hypothesis testing for t-test
	22/6/24	Saturday	Holiday
	24/6/24	Monday	Two sample t test

4th WEEK	DATE	DAY	NAME OF THE TOPIC/MODULE COMPLETED
	25/6/24	Tuesday	Simple Linear Regression
	26/6/24	Wednesday	Simple Linear Regression Project
	27/6/24	Thursday	Multiple Linear Regression
	28/6/24	Friday	Multiple Linear Regression continued
	29/6/24	Saturday	Holiday
	1/7/24	Monday	Polynomial and Exponential Regression

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1. EXECUTIVE SUMMARY

This report is about my 4 weeks internship program with **SkillDzire**. In this comprehensive report, I have discussed about every major aspect of the company which I observed and perceived during my internship program.

During my internship program, I have learned and mainly worked on Building Models from Training Data Sets. All the details have been discussed in detail. All the policies and procedures of the company have been discussed in detail.

As the main purpose of the internship is to learn by working in practical environment and to apply the knowledge acquired during the studies in real world scenario in order to tackle the problems using the knowledge and skill learned during the academic process.

2. OVERVIEW OF ORGANIZATION

SkillDzire is an organization focused on providing skill-based training and development, aiming to bridge the gap between academic knowledge and industry demands. They offer a widerange of training programs and internships tailored to equip students and professionals with practical, in-demand skills across various fields, including data science, software development, digital marketing, and business analytics.

Key Aspects of SkillDzire:

1. **Practical Training:** Emphasis on hands-on learning through real-world projects, simulations, and interactive workshops, allowing learners to gain practical experience.
2. **Industry-Relevant Curriculum:** Programs are designed to meet current industry standards, ensuring that participants acquire skills that are directly applicable to real-world scenarios.
3. **Expert Mentorship:** Experienced professionals guide learners, offering mentorship and insights to help participants better understand and navigate their chosen fields.
4. **Career Support:** SkillDzire provides resources and guidance for career advancement, including resume building, interview preparation, and networking opportunities with industry professionals.
5. **Flexible Learning:** Offers both online and offline training modules to cater to different learning preferences and schedules.

3. INTERNSHIP PART:

During an internship with Andhra Pradesh State Council of Higher Education (APSCHE) in the domain of Machine Learning (ML), an intern's activities and responsibilities typically include hands-on tasks, research, and collaboration. Here's a structured description:

1. Working Conditions:

- The working environment is typically a combination of virtual and on-site participation (if applicable).
- Interns work in collaboration with a mentor or a senior data scientist.
- The environment encourages independent research and development, as well as team work on specific projects.

2. Weekly Work Schedule:

- The weekly schedule generally consists of around 35-40 hours per week, depending on the scope of the internship.
- Daily work includes both learning (through tutorials, webinars, and reading) and implementation (writing code, building models).
- Weekly meetings and progress reviews are conducted with supervisors or project managers to track progress and discuss challenges.
- The schedule typically consists of:
 - Monday-Friday: Main workdays, dedicated to coding, testing models, and attending mentorship sessions.
 - Saturday: In some cases, dedicated to self-learning, documentation, or catching up on missed tasks.

- Sunday: Off or reserved for self-paced learning if required.

3. Tasks Performed:

- Data Collection and Preprocessing: Interns are involved in collecting, cleaning, and organizing datasets to make them suitable for training models.
- Use of libraries like Pandas and Numpy for data manipulation
- Model Training and Development: Developing machine learning models based on real-world problems such as classification, regression, and clustering using algorithms like Decision Trees, Random Forests, Neural Networks, and Support Vector Machines.
 - Use of frameworks like TensorFlow, Keras, and Scikit-learn for building and testing models.
- Model Evaluation: Interns are responsible for evaluating the performance of models using metrics such as accuracy, precision, recall, and confusion matrices.
 - They learn to fine-tune models by adjusting hyper parameters to improve model accuracy.
- Research and Implementation: Exploring and implementing new algorithms or optimization techniques to solve specific problems.
- Documentation: Interns maintain detailed documentation of their process, challenges faced ,and the solutions implemented during the internship.

4. Skills Acquired:

- Technical Skills:
 - Proficiency in Python programming and key ML libraries (TensorFlow, Keras, Scikit-learn).
 - Hands-on experience with data preprocessing techniques and model evaluation methods.
 - Understanding of core machine learning algorithms like regression, classification, and clustering.
 - Experience with deep learning models (e.g., Convolutional Neural Networks for image recognition).
 - Proficiency in using cloud-based tools for model deployment and testing.
- Soft Skills:
 - Time management and working on deadlines.
 - Collaboration and teamwork, especially in a virtual or hybrid setting.
 - Communication skills, particularly in explaining technical concepts to a non-technical audience.

4. OUTCOME DESCRIPTION:

During the SkillDzire Data Science Internship, the work environment was collaborative, structured, and supportive, fostering both professional and personal growth. Here's a detailed breakdown of various aspects:

1. People Interactions & Teamwork:

The environment promoted a collaborative spirit, with frequent interactions between interns, mentors, and team leaders.

There was an open communication culture, where interns could ask questions and seek guidance, fostering mutual support and teamwork.

Regular team meetings encouraged brainstorming, sharing of ideas, and collective problem-solving.

2. Clarity of Job Roles:

Roles and responsibilities were clearly defined from the outset, ensuring that each intern knew their tasks for the week or project.

Mentors provided specific guidance on projects and offered regular feedback, which helped maintain clarity in expectations.

3. Protocols, Procedures, and Processes:

Interns followed a structured workflow, which included data preprocessing, model building, evaluation, and documentation.

Standard operating procedures (SOPs) for coding, version control, and reporting were in place to ensure consistency and discipline.

There was a strong focus on adhering to best practices, such as maintaining code documentation and following established evaluation protocols for AI models.

4. Discipline & Time Management:

Interns were expected to follow a strict schedule with daily and weekly tasks assigned, promoting effective time management.

The work environment encouraged punctuality in attending sessions, submitting tasks, and participating in meetings, fostering a disciplined approach to work.

5. Harmonious Relationships & Socialization:

The workplace culture was inclusive and friendly, with opportunities for socialization through group activities and team-building exercises.

The atmosphere promoted harmonious relationships among interns, and collaboration was encouraged, with each individual feeling valued.

Informal communication channels, such as Slack or WhatsApp groups, helped in easy socialization and creating a sense of community.

6. Motivation & Mutual Support:

Mentors and team leaders were encouraging, providing positive reinforcement and constructive feedback to keep interns motivated.

Peer support played a significant role in completing tasks efficiently, with team members often helping each other overcome technical challenges.

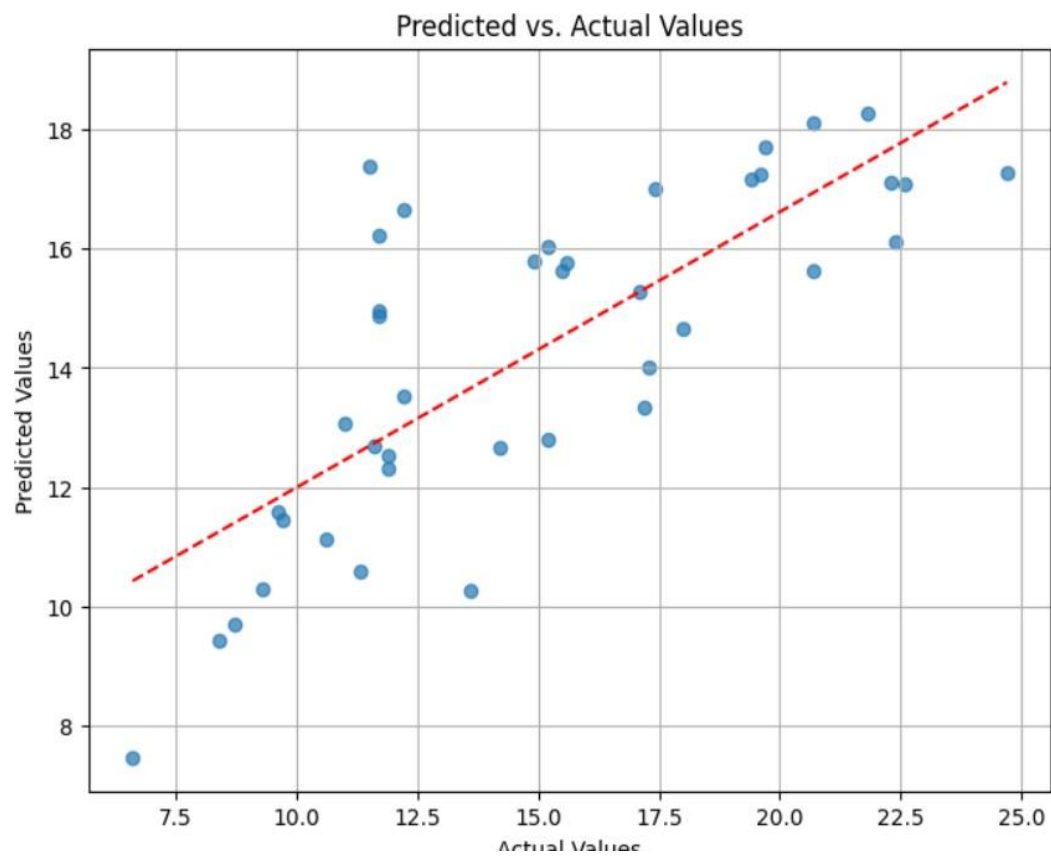
Group discussions, hackathons, and workshops further boosted learning and motivation.

7. Facilities Available & Maintenance:

The virtual or on-site facilities (depending on the mode of internship) were well-maintained, providing access to necessary tools, software, and cloud resources for work. Access to high-performance computing resources such as **Google Colab** or AWS for model training ensured a smooth work experience.

5. OUTPUTS:

	TV	Sales
30	292.9	21.4
108	13.1	5.3
68	237.4	18.9
4	180.8	12.9
138	43.0	9.6



6. CONCLUSIONS:

Continuous monitoring and responsive actions based on customer feedback will help the restaurant maintain a positive reputation and drive business growth.

Summarize the Sentiment analysis of restaurant reviews is a powerful tool for understanding customer satisfaction and identifying areas for improvement.

In this internship, we explored the application of machine learning techniques in sentiment analysis, focusing on how these tools can extract insights from text data. Through the implementation of various algorithms, including logistic regression, decision trees, and neural networks, we achieved significant accuracy in classifying sentiments expressed in the data sets.

The project highlighted several key takeaways:

1. **Data Preprocessing:** Effective preprocessing techniques, such as tokenization, stemming, and removal of stop words, were crucial in enhancing model performance.
2. **Model Selection:** Experimenting with different machine learning models allowed us to understand their strengths and weaknesses, enabling us to choose the most effective approach for our specific data set.
3. **Evaluation Metrics:** Utilizing metrics such as accuracy, precision, recall, and F1-score provided a comprehensive view of model performance, guiding us in optimizing our approaches.
4. **Real-World Applications:** The insights gained from sentiment analysis can be applied in various domains, including marketing, customer service, and social media monitoring, demonstrating the relevance and importance of this field in today's data-driven environment.

7. CHALLENGES FACED:

- At the beginning of internship, I faced difficulty for understanding the applications and different tools.
- I faced difficulty in installing the software.
- I faced difficulty in managing college and internship timings.
- I faced difficulty in understanding the advanced topics in android.
- I faced difficulty in managing the memory in pc.
- Even with these difficulties, I am able to complete the internship and it helps me in securing a new job.